ASSIGNMENT TWO

WEB PUBLISHING PROSOSAL AND FUNCTIONING HTML PROTOTYPE

CASE ORGANISATION: MAMBA COMMUNITY

Vincent Korir Enrollment NoBSCIT262710765

Introduction

This proposal mostly responds to the assignment case of a Kenyan community-based organisation whose existing website is difficult to navigate, inconsistently formatted, weak in alternative text and unclear in its presentation of programme data. The proposed Mamba Community website is designed for citizens, donors and partner institutions, including people using mobile devices, low-bandwidth connections and other assistive technologies. The website prototype is supplied as a separate folder containing three linked HTML pages, a shared stylesheet and two lightweight local SVG illustrations.

Important data note: Mamba Community is used as the case organisation name. No verified organisational programme statistics were supplied. Therefore, all figures in the prototype are explicitly labelled as illustrative academic sample data and must not be represented as actual results.

1.Analysis of the web publishing lifecycle

Web publishing lifecycle is a planned sequence through which a website is researched, organised, built, checked, released and maintained. The stages are connected: feedback and evidence gathered during testing or maintenance may require earlier design decisions to be revised before.

1.1 Requirements gathering

Meeting the organisational representatives and intended users to identify the website's purpose, content, functionality, constraints and success mitigation. For Mamba Community, essential requirements include programme descriptions, project results, contact details, downloadable resources, mobile support and accessibility. The team should always confirm who approves content, who updates it and which information to be remain private.

2. Audience Analysis

Identifying most of the needs, abilities, devices and connectivity conditions of citizens, donors, partner institutions and people using assistive technologies. Citizens may need simple explanations and contact details; donors may look for transparent results; partners may need programme documents. Some users may have limited data bundles, small screens, low vision or keyboard-only access. These requires justification of plain language, short pages, clear headings, readable contrast, meaningful alt text and low-bandwidth assets.

3. Information Architecture

Grouping information into a logical hierarchy and definition of page labels, navigation and relationships. The proposed top-level pages are Home, Programmes and Results. Contact information is consistently available in the footer. This reduces the effort needed to locate information and avoids placing unrelated programme data in long unnecessary paragraphs.

4. Prototyping

Creating a sitemap, wireframes and a clickable HTML prototype before full release. We Ask representative users to complete tasks such as finding a programme, locating a result and finding contact details. Revising labels, page order and layout when users hesitate or misunderstand them.

5. Implementation

Building with semantic HTML and a shared CSS stylesheet. Using headings, landmarks, descriptive links, lists, a properly marked-up data table and useful image alternatives and most important keeping the content separate from presentation. I Avoid deprecated presentation attributes and use CSS for layout and styling. Local lightweight SVG illustrations reduces reliance on external image services.

2. Proposed site structure and navigation model

The prototype uses a shallow hierarchy with three primary pages. The same main navigation appears on every page, using descriptive labels and a current-page indicator. A persistent footer contains the organisation's contact email. This gives users predictable navigation and avoids forcing them to learn a different menu on each page.

Sitemap:

Mamba Community website

- Home (purpose, overview and quick links)

- Programmes (education and skills; health awareness; environment and livelihoods)
- Results (illustrative programme-results table and data-quality note)
- Contact email is available in the footer on every page

Page hierarchy formatting and content grouping: Home introduces the organisation and directs users to priority information. Programmes groups related activities under clear headings. Results present comparable.

programme outputs in a table contain Contact details that remain consistently available in the footer. Internal hyperlinks connect the three pages and relevant sections. The email link is an external application link using the mailto scheme; before publication it must be replaced with the organisation's verified address. Links should use descriptive text such as “View programme results”, rather than vague wording such as “click here”.

Lists: An ordered list is an appropriate for activities that follow a sequence, such as identifying needs, organising workshops and collecting feedback. An unordered list is suitable for non-sequential navigation or quick information links. A definition list is appropriate for terms such as community-based organisation, monitoring and outcome because each term is paired with its explanation. Lists improve scanning and expose relationships to technologies.

3. Standards-aware HTML prototype

Separating the website folder contains index.html, programmes.html and results.html, styles.css and two SVG images in an assets folder. Each page starts with the HTML5 doctype and includes the html, head, title and body elements. The lang="en" attribute identifies the main language, while the viewport meta tag supports mobile rendering.

Semantic structure: header, nav, main, section, article and footer elements identify the purpose of page regions. A skip link lets keyboard users bypass repeated navigation. Heading levels are used to organise content. Descriptive link text communicates destination. The shared stylesheet separates presentation from document structure and makes visual changes easier to maintain consistently.

Images and alternative text: two locally stored SVG illustrations are included. Their alt attributes describe the meaningful content of each image. If an image is purely decorative in a future version, it should use an empty alt attribute (alt="") so assistive technologies can ignore it. These prototype does not rely on remote image hosts, reducing the chance of missing assets and avoiding unnecessary third-party requests.

Emphasis: strong is used for important warnings, such as the fact that the data is illustrative. Emphasis should communicate meaning rather than merely create visual decoration. CSS

controls colours, spacing, responsive columns and focus appearance. Deprecated presentation attributes such as bgcolor, align and font should not be used; CSS provides a more maintainable approach.

The complete source code is provided in the accompanying website ZIP. Open index.html in a browser and use the shared navigation to reach the other pages.

4. Accessible programme-results table

The Results page contains a data table with a caption; column headers marked with scope="col" and row headers marked with scope="row". It represents four illustrative programme outputs: digital literacy participants, health awareness sessions, trees planted and youth workshop completions. The caption explains the table's subject and reporting period concept; a note explicitly states that the figures are fictional sample data**.**

Programme	Activity	Target	Illustrative achievement	Progress
Education	**Digital literacy participant's**	**200**	**96**	**80%**
Health Awareness	**Community Awareness sessions**	**30**	**25**	**73%**
Environment	**Trees Planted**	**600**	**425**	**95%**
Youth skills	**Young people workshops**	**80**	**68**	**85%**

These table is appropriate when information has genuine row-and-column relationships and users needed to compare values across categories. It is not appropriate merely to position page elements. Using a table for layout can create confusing reading order and unnecessary cell navigation for screen-reader users. CSS grid or flexbox should be used for visual layout instead.

5. How to open and use the prototype

1. Download Mamba_Community_HTML_Prototype.zip.

2. On Windows, right-click the ZIP file and choose Extract All.

3. Open the extracted folder and locate index.html.

4. Double-click index.html or right-click it and open it with Chrome, Microsoft Edge or Firefox.

5. Use the navigation at the top of the page to open Programmes and Results.

6. Keep styles.css and the assets folder in their original locations. Moving or renaming them may break styling or image paths.

The site is a static prototype and can be viewed locally without a web server.

6. Evaluation, risks and improvements

The prototype addresses several case's known weaknesses, but a prototype is not proof of full WCAG conformance or production readiness. The following risks should be tested and improved before launch.

- **Accessibility and assistive technology**

Risk: SVG alternative text, contrast, heading order or keyboard interaction may not work equally well with every browser and screen reader. Improvements: validate markup, check contrast against WCAG 2.2 AA targets, test all links by keyboard, confirm visible focus and conduct screen-reader checks with NVDA or VoiceOver. Ask users with disabilities to test key tasks.

- **Responsive design and small screens**

Risk: navigation or tables may become cramped on narrow screens or at high zoom. Improvement: test at 320px, 375px, tablet and desktop widths; use responsive CSS, allow wrapping, avoid fixed-width page containers and provide a deliberate mobile table treatment such as horizontal scrolling with an accessible label if needed.

- **Performance and low bandwidth**

Risk: large images, unnecessary scripts or external dependencies could make pages slow and expensive to load. Improvement: keep assets small, compress photographs, use responsive image sources for production photos, minimise scripts and test using throttled network settings and Lighthouse/PageSpeed Insights. The prototype uses small local SVGs and no JavaScript framework**.**

- **Navigation and link reliability**

Risk: users may encounter broken links, ambiguous labels or missing images. Improvement: run automated link/path checks, manually test every navigation link and email link, use descriptive labels and recheck links after deployment.

- **Security, privacy and maintenance**

Risk: contact forms or published documents may expose personal information or become outdated. Improvement: collect only necessary information, protect any form submissions, restrict editing access, use HTTPS and backups, appoint content owners and schedule regular reviews.

- **Testing plan and report template**

checks are specified for the prototype. The HTML package can be inspected locally; browser and assistive-technology results must be completed on the actual device used for submission. Do not report a browser or screen-reader test as passed unless it has been performed.

Test	Method/Expected result	Results to Record
HTML Structure	Running an Html validator, no major parsing errors	Completing after running validator
Internal links and assets	Opening all three pages by confirming each link And svg path	Completing after local browser checks

Keyboard access	Using Tab , shift+Tab and Enter, focusing on visible and skip link	Completing on device
Browser Compatibility	Testing Current chrome and Firefox or edge, navigation and layout remaining usable	Completing on device
Accessibility	Checking contrast and testing with a screen reader a, header are announced meaningfully	Completing with selected tools

Conclusion

The proposed Mamba Community website responds to this case by organising information into a simple three-page structure, using semantic HTML, providing image alternatives, presenting programme outputs in a real data table and applying responsive styling. The lifecycle approach ensures that the website is shaped by audience needs and maintained after being launched. Before publication, the organisation must replace illustrative statistics and placeholder contacts details with verified, researched and approved information and complete the proposed testing.

References

World Wide Web Consortium (W3C). (2024). Web Content Accessibility Guidelines (WCAG) 2.2. https://www.w3.org/TR/WCAG22/

Supports accessibility requirements including text alternatives and perceivable, operable, understandable content.

MDN Web Docs. HTML: A good basis for accessibility. https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/Accessibility/HTML

Supports semantic HTML, descriptive links, alternative text, keyboard accessibility and accessible table headings.

MDN Web Docs. Responsive web design. https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/CSS_layout/Responsive_Design

Supports fluid layouts, media queries and responsive media.

web.dev. Serve responsive images. https://web.dev/articles/serve-responsive-images

Supports selecting appropriately sized images to reduce data use and improve loading performance

Appendix A: Prototype file inventory

- index.html: home page
- programmes.html: programme descriptions, ordered list and definition list
- results.html: accessible illustrative data table
- styles.css: shared responsive presentation
- assets/learning-community.svg: learning illustration
- assets/tree-planting.svg: environmental action illustration
- README.txt: opening and data notes